Understanding Mathematical Concepts via Go Language

Real Numbers

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Real Numbers

Axioms of Real Numbers

1. Addition

$$x + y = y + x$$

2. Multiplication

$$x \cdot y = y \cdot x$$

3. Ordering

$$x < y$$

4. Completeness

$$\text{Every Cauchy sequence converges.}$$

Real interface

```go
type Additive interface {
        Add(Additive) Additive
        AddInv() Additive
        Zero() Additive
}

type Multiplicative interface {
        Mul(Multiplicative) Multiplicative
        MulInv() Multiplicative
        One() Multiplicative
}
```

```go
type Element interface {
	Equals(any) bool
}

type Real interface {
	Element
	Additive
	Multiplicative
	GreaterThan(Real) bool
	ToFloat() float64
}
```

Integer

```go
func Integer(f Real, n int) Real {
	if n == 0 {
		return f.Zero().(Real)
	}
	o := f.One().(Real)
	if n < 0 {
		o = o.AddInv().(Real)
	}
	for i := 0; i < n; i++ {
		o = o.Add(o).(Real)
	}
	return o
}
```

Rational

```go
func Rational(f Real, r [2]int) Real {
	if r[1] == 0 {
		return nil
	}
	if r[0] == 0 {
		return f.Zero().(Real)
	}
	a, b := Integer(f, r[0]), Integer(f, r[1])
	return a.Mul(b.MulInv()).(Real)
}
```

Sqrt Function

$$a_{n+1} = \frac{1}{2}\left(a_n + \frac{x}{a_n}\right), \quad a_n \to \sqrt{x}$$

```go
func Sqrt(a Real) Real {
	x := a.One().(Real)
	half := Rational(x, [2]int{1, 2})
	for !a.Equals(x.Mul(x).(Real)) {
		x = x.Add(a.Mul(x.MulInv()).(Real)).(Real)
		x = x.Mul(half).(Real)
	}
	return x
}
```

Implementation

```go
type SimpleReal float64

const epsilon = 1e-6

func (f SimpleReal) Equals(x any) bool {
	if xfloat, ok := x.(SimpleReal); ok {
		return f-xfloat > -epsilon && f-xfloat < epsilon
	}
	return false
}

func (f SimpleReal) GreaterThan(g gocalc.Real) bool {
	return f.ToFloat() > g.ToFloat()
}

func (f SimpleReal) ToFloat() float64 { return float64(f) }
```

```go
func (f SimpleReal) Zero() gocalc.Additive { return SimpleReal(0) }

func (f SimpleReal) Add(g gocalc.Additive) gocalc.Additive {
	return SimpleReal(float64(f) + g.(SimpleReal).ToFloat())
}

func (f SimpleReal) AddInv() gocalc.Additive { return SimpleReal(-f) }

func (f SimpleReal) One() gocalc.Multiplicative { return SimpleReal(1) }

func (f SimpleReal) Mul(g gocalc.Multiplicative) gocalc.Multiplicative {
	return SimpleReal(float64(f) * g.(gocalc.Real).ToFloat())
}

func (f SimpleReal) MulInv() gocalc.Multiplicative {
	if f.Equals(f.Zero()) {
		return nil
	}
	return SimpleReal(1 / f)
}
```

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